

Engineering Physics

BRANCH: COMMON / BASIC SCIENCES

COURSE CODE: PH101

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for Engineering Physics (PH101) under the Common / Basic Sciences department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: Wave Optics

Interference: thin films, Newton's rings. Diffraction: Fraunhofer and Fresnel, single slit, double slit, diffraction grating. Polarization: double refraction, Nicol prism.

Unit 2: Quantum Mechanics

Inadequacy of classical mechanics. Wave-particle duality, de Broglie hypothesis, Heisenberg uncertainty principle. Schrodinger wave equation (time-dependent and independent), particle in a 1D box.

Unit 3: Lasers & Fiber Optics

Einstein coefficients, population inversion, Ruby and He-Ne lasers. Fiber Optics: principle, numerical aperture, single-mode and multi-mode fibers.

Unit 4: Semiconductor Physics

Band theory of solids, Fermi-Dirac distribution, intrinsic and extrinsic semiconductors, carrier transport, Hall Effect.

Uncertainty principle: $\Delta x \cdot \Delta p \geq \frac{h}{2}$. Numerical Aperture (NA) = $\sqrt{n_1^2 - n_2^2}$.

CRITICAL FORMULA / PRINCIPLE: Key Formulas & Exam Pointers

